

REPORT ON COVID-19 DATA

In this report I am focusing on the Covid-19 World Data with regards to the number of Confirmed Cases and Deaths from Covid-19. This report is based on clustering different regions of the world into groups based on the similarities they have in the number of Confirmed Cases and Deaths from Covid-19 in order to assess and compare the countries which belong to Very High , High, Medium and low risk Regions.

The report is based on the analysis done on the Covid-19 World Data and it is to assess and group Regions according to the data using two features which are the number of Confirmed Cases and Deaths from covid-19 across different countries to know which Regions were at Very High , High, Medium and low risk Regions in January 2021 before the use of the covid-19 Approved Vaccines and if their status change in January 2022 after the using of the approved Covid-19 Vaccines.

The data contains some rows where the Province/State is null. For those rows the value in Countries/Regions were used as Province/State, therefore from the preprocessing we have 201 unique countries and 773 unique states

During the period January 2021, 3405 states and regions were involved in the clustering while in January 2022, 3249 states and regions were involved in the clustering. Both clusters are based on two features which are the number of Confirmed Cases and Death from Covid19.

From the above I got my cluster data and used the Scaler.fit model to normalise the data and after which I performed the clustering algorithm on the normalised data.

I cluster the region into

- Very High risk Region (coloured Darkred spots in the map)
- High risk Region (coloured Blue spots in the map)
- Medium risk Region (coloured Yellow spots in the map)
- Low risk Region (Coloured Green spots in the map)

The result from my algorithm analysis shows in January 2021 Regions in countries like France , England, turkey were Very High Risk Region affected by the covid 19 pandemic based on the two features above.

Also some regions like Trinidad and Tobago, British island in South America regions were also High Risk Region with affected by the covid-19. Meanwhile, regions in Madrid, Poland, Lima along the coastal line to Santiago, Japan, morocco California, Florida were Medium Risk Regions but with high numbers which were not too much affected by covid-19. In addition 3371 regions out of 3405 regions were very low risk regions not very affected regions were in Africa, America, and northern Europe countries

According my analysis of this same regions in January 2022, only France remain in the Very High Risk Region with over 300percent increased confirm cases and deaths. Regions in England, Poland, Argentina, turkey became a High Risk Region, While Florida became the only very low risk region In January 2022. On the other 3241 regions out of 3249 regions became medium risk regions not very affected regions according to my algorithm with very low figures

According to forbes (2021) President Biden of the United State described the United Kingdom, some Schengen Countries, Brazil as High Risk Regions and will not allow anyone coming from these regions into the United States, this statement support my finding from January 2021 which group England which is in the United Kingdom, France which is in the Schengen nation as Very High Risk Regions based on the number of Confirmed Cases and Deaths

According Who Health Organization (2023), France covid-19 situation worsen in January 2022 with over 2, 427,055 confirmed cases with a weekly 414,044 rise which support my finding that France remained in the Very High Risk Region in 2022 January.

According to the CDC and World Health Organization the covid-19 affected regions increase with the confirmed cases and death increasing also. This also follows my finding which shows that most of the low risk region in January 2021 becoming a medium risk region which implies their confirmed cases and death increased.

However, some countries like Spain, Italy were not spotted in my analysis vividly which were areas and regions which were conspicuously affected by the covid-19 pandemic in these periods.

Below are the maps of the January 2021 and January 2022 Covid Affected Regions

Figure 1 2021 January Covid !9 Region Map



Figure 2 2022 January Covid !9 Region Map



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